

Factor out the greatest common factor (GCF)

Algebra I | CK-12: Factoring when the Leading Coefficient Equals 1

Key Idea

The **GCF** is the largest factor every term shares — both the number and the variables. Factoring it out is the reverse of distributing.

Key Terms

- **GCF** — greatest common factor
- **Factor** — rewrite as a product

Worked steps

Step 1: Find the GCF of the numbers and the smallest power of each variable.

Step 2: Divide each term by the GCF and write GCF(quotient).

Example: $6x^2 + 9x$: GCF is $3x$, so $6x^2 + 9x = 3x(2x + 3)$.

You Try

Factor $10x + 15$.